



CONTACT

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TECHNICAL SKILLS

Programming

C C++ Java Python

Embedded C/C++ on a microcontroller, written from scratch (robot control kernel)

Real-time loops, sensor reading, motor control, state machines

Debugging and tuning on real hardware

Engineering foundations

Mechanics and statics (forces, moments, equilibrium)

Thermodynamics and heat transfer basics

Calculus, linear algebra, proof-based maths

Learning now

CAD (Fusion 360) FEA
Radio / RF ECSS basics

AMATEUR RADIO

Preparing the **ANACOM Category 1** amateur radio licence exam (Porto, October 2026). Goal: exchange text messages through the geostationary satellite **QO-100** (2.4 GHz up, 10.5 GHz down). Radio theory, antennas and link budgets feed directly into the ICARUS ground station and UHF link.

SOFT SKILLS (WITH PROOF)

Teamwork: 4-person robotics team, national team competitions

Resilience: 4 consecutive years of merit awards

Responsibility: district badminton referee, 2025 and 2026

Adaptability: Erasmus+ exchanges, Poland 2022, Serbia 2023

LANGUAGES

Portuguese: native
English: upper intermediate (B2)

Miguel Brito

Civil Engineering student, FEUP

Applicant, Porto Space Team, ICARUS CubeSat: Structures & Thermal Control (1st choice) and On-Board Computer

PROFILE

First-year engineering student with top marks in Mathematics and Physics (20/20) and hands-on embedded experience: I wrote the control kernel of a competition robot from scratch in C/C++ and helped my team win **2nd place at BotOlympics 2026**, sponsored by Critical Software. I want to apply structural mechanics and careful, tested code to a satellite that flies, and to learn CAD, FEA and flight software standards along the way.

EDUCATION

BSc Civil Engineering, 1st year

Sep 2026 to present

Faculty of Engineering, University of Porto (FEUP)

- Core path in statics, mechanics of materials and structural analysis: the same tools used to size a CubeSat chassis and its deployable mechanisms against launch loads.

Secondary education, Science and Technology

2023 to 2026

Escola Secundária Avelar Brotero, Coimbra. Final average **18.5/20**

- **Mathematics 20/20:** calculus and algebra behind FEA, thermal models and attitude estimation.
- **Physics 20/20:** mechanics, thermodynamics and electricity, the physics of loads, heat balance and power budgets.
- High merit award every secondary year (2023/24, 2024/25, 2025/26).

PROJECTS AND ACHIEVEMENTS

BotOlympics 2026: 2nd place (team Prode-ESAB)

Mar 2026, Coimbra

Robotics competition by NEEEC, University of Coimbra. Gold sponsor: Critical Software

- Wrote the robot's control kernel from scratch in C/C++: sensor polling, motor control and a navigation state machine for an autonomous maze delivery task.
- Tuned the code between runs under time pressure; placed 2nd in the public final.
- *Learned: on embedded hardware every byte and millisecond counts, and only a test run proves the design. Directly relevant to OBC flight software.*

PmatE National Science Competition: 1st place, Mathematics (team)

2026, Univ. Aveiro

- Won a University of Aveiro study grant with my team. *Learned: peer checking catches errors faster than redoing work, the core of design reviews.*

Mathematics olympiads and training

2019 to 2026

- National finals, Portuguese Mathematical Olympiad (2024). Projeto Delfos olympiad training, University of Coimbra (2024 to 2026).
- Canguru Matemático: 3rd at school level (2019), 1st at school cluster level (2021).
- *Learned: estimate first, then prove; discard wrong paths early. The habit behind sanity-checking any structural or thermal result.*

Programming tutoring: C, C++ and Java

2024 to 2025

- Private tutoring beyond the curriculum: memory and pointers in C, object-oriented design in C++ and Java. Put into practice at BotOlympics.

Academic recognition

2022 to 2023

- Merit grant, Parish of Coimbra (2022/23). Distinguished student, 9th grade (2023).

WHY I FIT ICARUS

Structures & Thermal Civil Engineering mechanics, Physics 20/20, eager to learn FEA, Esatan-TMS and the ground antenna build.

On-Board Computer Proven embedded C/C++ from scratch; motivated by memory-constrained, reliable code.

Communications Amateur radio licence in progress; ICARUS uses the amateur UHF band.